

Dinesh Kovirineni

AI Software Engineer | Backend & Cloud Engineering

(703) 581-2813 • [Email](mailto:dineshkovirineni0924@gmail.com) (dineshkovirineni0924@gmail.com) • [LinkedIn](https://www.linkedin.com/in/dinesh-kovirineni) (linkedin.com/in/dinesh-kovirineni) • [GitHub](https://github.com/dineshkovirineni) (github.com/dineshkovirineni) • [Portfolio](https://dineshkovirineni.github.io/Dinesh-Portfolio/) (dineshkovirineni.github.io/Dinesh-Portfolio/)

Profile Summary

AI Software Engineer with 3+ years of experience building AI-powered applications, Generative AI solutions, scalable backend services, and cloud-native systems. Experienced in LLM applications, RAG, semantic search, vector embeddings, and AI-driven catalog intelligence using Python, FastAPI, Vertex AI, Pinecone, LangChain, and GCP. Strong background in microservices, REST, GraphQL, gRPC, event-driven systems, Go, Java, Kafka, Kubernetes, Docker, AWS, and GCP.

Technical Skills

Languages: Python, Go, Java, Elixir, TypeScript, JavaScript
AI & ML: Generative AI, LLM Applications, RAG, LangChain, Vertex AI, Hugging Face, OpenAI, Vector Embeddings, Semantic Search, NLP, Prompt Engineering
Backend: FastAPI, REST APIs, GraphQL, gRPC, Microservices, Event-Driven Architecture, Distributed Systems
Databases: Pinecone, Cassandra, PostgreSQL, MySQL, BigQuery, Redis, MariaDB
Cloud & DevOps: GCP, AWS, GKE, Kubernetes, Docker, Kafka, Jenkins, CodePipeline, CloudWatch, CloudFormation, Git

Work Experience

Software Engineer, AI & Merchandising (Contract) Sam's Club (Walmart) <u>Catalog AI</u> - Developed an AI-powered catalog intelligence platform using React, TypeScript, Python, FastAPI, and GKE for automated product categorization and semantic search. - Built RAG workflows with Vertex AI and Pinecone to improve product-to-category matching and reduce manual catalog curation. - Engineered vector embedding pipelines using Vertex AI and Cloud Storage for SKU similarity matching and automated category recommendations. - Implemented event-driven catalog ingestion using Kafka and GCP Pub/Sub for real-time product updates. <u>Pricing RT</u> - Built Go-based ingestion services to consume pricing feeds and synchronize data with Walmart eCommerce schemas through schema transformation. - Developed Java gRPC services with Redis caching, reducing pricing query latency by 50%. - Enabled real-time pricing synchronization using Kafka and Cassandra, replacing batch processing. <u>In-Club Checkout Modernization</u> - Migrated in-club checkout systems from legacy SMART architecture to Elixir-based cloud architecture using GCP and Kubernetes across 600+ club locations. - Built high-throughput Go microservices and GraphQL services for CPC document ingestion and real-time POS APIs, reducing query latency to under 100ms and improving retrieval efficiency by 35%.	May 2024 – Present Frisco, TX
Software Engineer EPSoft	May 2023 – May 2024 Texas, USA
- Developed scalable Python/FastAPI backend microservices using AWS EC2, Lambda, and S3 for high-throughput API workloads. - Designed RESTful APIs and implemented CloudWatch and CloudFormation for monitoring and infrastructure management. - Built CI/CD pipelines using Jenkins and AWS CodePipeline; containerized services with Docker and deployed applications in Kubernetes.	

Academic Projects

Text-to-SQL Analytics Agent | [GitHub](#)

- Built a LangGraph-orchestrated AI agent that converts natural-language questions into validated BigQuery SQL — with a multi-stage pipeline: semantic schema retrieval (pruning 7 tables to the relevant subset and closing join paths), AST-level guardrail validation, cost dry-run against BigQuery, and a self-correction loop that feeds exact errors back to the model, achieving 85% first-try query success across 50 test questions.
- Implemented 7 AST guardrail checks via sqlglot — blocking DML/DDI, hallucinated columns, unknown tables, stacked-statement injection, and unbounded queries — before any query reaches the warehouse; added cost control with dollar-estimate gating and a BigQuery byte ceiling as a second line of defense.
- Shipped a FastAPI HTTP layer, a Dockerfile with an offline SQLite mirror (no credentials required), and a 172-test CI suite covering guardrails, schema retrieval, repair logic, and full pipeline smoke tests across Python 3.10–3.12.

Tools: Python · LangChain · LangGraph · Claude · GCP BigQuery · FastAPI · sqlglot · Docker

Automated Document Summarization & Extraction Tool | [GitHub](#)

- Built a RAG pipeline that ingests PDF, DOCX, TXT, and MD files into a Pinecone serverless index (token-aware chunking with namespace isolation per document), then runs six focused per-field-group extraction passes using a fine-tuned Hugging Face LLM — achieving 92% field-extraction accuracy against gold labels.
- Implemented self-consistency voting (K passes per field group, majority merge) and evidence grounding (extracted strings matched back against source chunks, with digit-presence penalties for hallucinated values) — weighted 65/35 grounding-over-vote-agreement — so fields below the confidence floor are dropped rather than reported.
- Delivered a Pydantic schema-salvage layer (a bad field is discarded, the rest kept), a map-reduce executive summary pipeline, a CLI evaluation harness with micro/macro accuracy and per-field breakdown, and 47 offline tests covering JSON recovery, voting, grounding, and schema salvage — reducing manual document review time by 45%.

Tools: Python · Hugging Face · Pinecone · Pydantic · pytest

Education

M.S. in Computer Science

George Mason University, Virginia, USA

May 2023

B.Tech. in Information Technology

Gokaraju Rangaraju Institute of Engineering and Technology, Telangana, India

August 2021